ENGR 102 Summer Bridge
Bridge Day 14 Student Workbook

Lists, Length, Indexes, Mutation, and List Loops

Thursday, July 23, 2026 • 50 points • tagged v5 successor

Name		UIN		Section	

Daily target and independence boundary

Choose index evidence when stored values must be read, changed, summarized, and located.

Allowed tools	Prior tools plus lists, len, append, indexed reads, and indexed assignment
Support supplied	A starting list and required final-state summaries are supplied.
Student decides	Loop form, list mutation, summary state, and first-largest tracking.
Scaffold level	Specification only; no planning prompts or variable inventory.

Evidence and scoring

Evidence	Points	Secure work shows
Integrated trace	10	intermediate state, condition results, exact output, and meaning
Diagnosis and repair	8	actual, intended, cause, repair, and a discriminating test
Complete program and tests	32	coherent executable logic, required output, assumptions, and defended tests

Task 1. Integrated trace - 10 points

Trace index state, list values, an appended evidence list, and an accumulator in one ledger.

```
values = [6, 3, 9, 4]
total = 0
low_positions = []

for index in range(len(values)):
if values[index] < 5:
low_positions.append(index)
else:
total = total + values[index]

print(total, low_positions, values[len(values) - 1])
```

Your task

1. Give the complete index-by-index ledger and exact output, distinguishing each index from its stored value.

Task 2. Diagnose and repair - 8 points

The intent is to count every negative value, including the first and last positions.

```
values = [-2, 4, -1, 6, -3]
negative_count = 0

for index in range(1, len(values) - 1):
if values[index] < 0:
negative_count = negative_count + 1

print(negative_count)
```

Your task

1. Give actual versus intended result, identify both skipped boundaries, repair the loop, and name one compact test that protects both ends.

Task 3. Construct a complete program - 32 points

Program specification

- Use readings = [12, -3, 7, 20, 7]. The list is nonempty.
- Loop by index and replace every negative value with 0.
- Using final values, compute valid_total and the number of values at or above 10.
- Find the index of the first largest final value without using max or index.
- Print the final list, valid_total, high_count, and largest_index on separate lines.

Program workspace

Task 3 continued. Test and defend - included in 32 points

Continue code if needed. Required tests, expected results, and brief defense

Bridge Day 14 VIP Evidence Handoff

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Why engineers use this page

Begin with the notebook's required input/conversion/f-string reconnect, then carry index, value, and final-list evidence into reverse traversal and one- or two-condition filters.

Decision boundary: Code is useful only when its stored state, visible result, assumptions, units, limits, and tests support a defensible engineering interpretation.

Construct readiness before independent work

New authoring tools: string traversal, list traversal, list index read, list index write, string index read, slice expression, membership test, parallel list indexing, compound filter
Carried authoring tools: assignment, print call, arithmetic expression, input call, int conversion, float conversion, f string, comparison expression, boolean operator, if elif else, for loop, range call, append call, len call
Supplied-code exposure only: none
An exposure-only construct may be traced in complete supplied code, but the independent task will not require students to invent it before its authoring day.

Notebook cells and task constructs

Activity	Work	Stable cells	Required authoring constructs
18.25	Reverse Values	`18-25-work / 18-25-transfer / 18-25-cold`	append call, for loop, range call
18.26	Values At Or Above Cutoff	`18-26-work / 18-26-transfer / 18-26-cold`	append call, comparison expression, for loop
18.27	Feasible Workstations	`18-27-work / 18-27-transfer / 18-27-cold`	append call, boolean operator, f string, float conversion, for loop, input call, int conversion, range call

Readiness evidence

1. State which mapped activity requires indexes and which activities can preserve order by traversing values.

2. For Activity 18.27, write the two Boolean constraints separately before combining them with 'and'.

Timing and help trigger

Record a first prediction before running. If you still lack an executable plan after about 8 focused minutes, use the linked ThinkCSPy refresher or the supplied model. If two attempts fail for the same named requirement, write the exact mismatch and ask a peer mentor or instructor a targeted question. Timing is diagnostic evidence, not a speed grade. **Start or elapsed minutes:** ____________ **My help trigger:** ____________

Engineering Evidence Record

Complete one record from a model, transfer, or Independent Program Studio build. Generic statements are not sufficient; use actual values, a real revision, and one concrete next test.

Evidence field	Record
Prediction	
Observed Result	
Revision Reason	
Engineering Meaning	
Units Or Not Applicable	
Assumption	
Model Limit	
Next Test	
Help Trigger	
Minutes Used	

Notebook and submission procedure

1. Attempt, predict, run, inspect, and revise before opening a reveal.
2. Complete the end-of-notebook Independent Program Studio without a reveal or copied solution. Only those visibly labeled Studio cells are scored for independent mastery.
3. Save or download the completed notebook as one `.ipynb` file.
4. Upload that one notebook to the matching Gradescope assignment. Do not export a `.py` file or create a ZIP.

Generated from the canonical VIP construct and evidence contract; workbook planning and executable notebook work remain distinct.